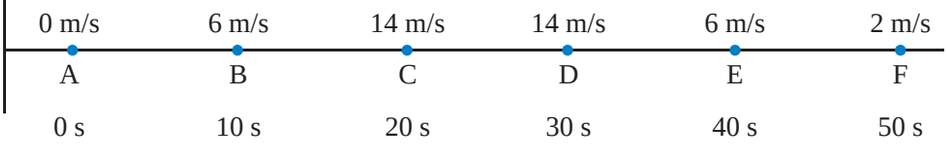
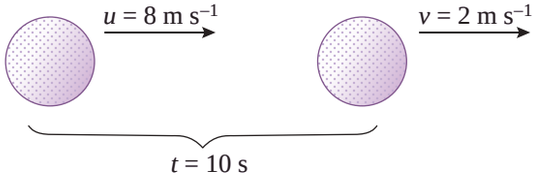
The velocity of a train at different times is given in the figure. Analyse this and complete the table .



The distance travelled by train	Initial velocity (u) m/s	Final velocity (v) m/s	Change in velocity (v – u) m/s	Time taken (t) s	Acceleration = Change in velocity / Time a = (v – u) / t m / s^2
A-B	0	6	6	10	0.6
B-C					
C-D					
D-E					
E-F					

2.3.2 Negative acceleration or Deceleration or Retardation

If the velocity of an object decreases with respect to time, then the object is said to be in negative acceleration or deceleration or retardation.



The velocity of a golf ball rolling in a straight line changes from 8 m/s to 2 m/s in 10 s. What is its deceleration, assuming that it is decelerating uniformly ?

Initial velocity (u) = 8 m/s

Final velocity (v) = 2 m/s

Time taken(t) = 10 s

Acceleration (a) $= \frac{(v - u)}{t} = \frac{(2 - 8)}{10} = -0.6\,m/s^2$

The deceleration is –0.6 m/s^2

2.3.3 Uniform acceleration

An object undergoes uniform acceleration when the change (increase or decrease) in its velocity for every unit of time is the same.

The table given below shows the uniform acceleration of a bus.

Time (s)	1	2	3	4	5
Velocity (m/s)	20+20	40+20	60+20	80+20	100 + 20
	(acceleration)				
Velocity (m/s)	100–20	80-20	60-20	40-20	20-20
	(deceleration)				

When the velocity of the object is increasing by 20 m/s the acceleration is 20 m/s^2. When the velocity of the object is decreasing by 20 m/s the deceleration is 20 m/s^2.

2.3.4 Non – uniform acceleration

An object undergoes non–uniform acceleration if the change in its velocity for every unit of time is not the same.

Time (s)	0	1	2	3	4	5
Velocity (m/s)	0	10	40	60	70	50
Change in Velocity (m/s)	0	10	30	20	10	20

Note here that the change in velocity is not the same for every second. Thus, the moving object is undergoing non-uniform acceleration.

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